

Education

BSc (Hons) Computer Science **2022-2025** **University Of Manchester**
Grade: First Class **Societies:** Computational Finance (Co-head), Computer Science, Mathematics, Radio
Dissertation: *Exploring the Semantics of Recursion in Register Machine Programs with Category Theory*
Courses: *Machine Learning, Compilers, GPU & Parallel Programming, Advanced Algorithms & Data Structures*

Experience

C++ Software Engineer **June 2025-**
TuringDB **London**

- Leading engineer on the world's fastest analytical graph database engine: vectorised and column-oriented; in C++20
- Facilitated **lock-free** read transactions via a Git-like DB version control system; which provided **snapshot isolation**
- **Reduced CPU cycle counts by 66%** on hot-path DB operations; motivated by **perf** and **callgrind** profiling
- Researched, designed, and implemented novel **cache-conscious** join algorithms; **50x+ faster** than competitors
- Improved all stages of **query interpretation**: parsing; syntactic and semantic analysis; optimisation; and execution
- Leveraged **data-orientated design** in implementing various **entity component systems** to match access patterns
- Lead **Nix** cross-platform, cross-architecture packaging and development/build environments for reproducibility

Freelance Python Software Engineer **2023-2025**
Optimistics **London**

Projects

Monte Carlo Tree Search Chess Agent <https://github.com/cyrusknopf/yggdrasil>

- Built a bitboard agent with my own implementation of the Monte Carlo tree search algorithm in modern **C++23**
- Followed **test-driven development** with Google Test, in a growing test suite of **over 100** unit tests
- Data-driven optimisations motivated by **microbenchmarking** with Google Benchmark resulting in **10x** speedups
- Identified hot-path and key functions to target optimisations using **profiling** tools such as Callgrind (Valgrind tool)

Mini Operating System for ARM microcontroller <https://github.com/cyrusknopf/arm-assembly-os>

- Engineered core OS functions, extending them to develop a program with multimodal I/O interaction with the user
- Added support for software debouncing with bespoke **interrupt handlers** and **supervisor calls**
- Synthesised **Verilog** onto an **FPGA** to uniquely extend functionality and facilitate more complex programs

PRISM - Portfolio Risk, Investment & Strategy Management <https://github.com/cyrusknopf/PRISM-hackathon>

- Orchestrated a university hackathon with **100+ participants** by designing a trading platform using **Go** and **Python**
- Engineered a challenge requiring **NLP** analysis and portfolio selection using an optimised, bespoke, self-hosted **LLM**
- Built a **containerised infrastructure** (Docker, Go, Python, Next.js, Postgres) handling **80,000+** requests per second
- Implemented a multi-metric evaluation system as a function of Sharpe, Sortino, and diversity for portfolio evaluation

Open Source Obsidian Plugin <https://github.com/cyrusknopf/obsidian-smart-tasklists>

- Contributed an **open-source** plugin to the Obsidian note-taking platform, written in **Typescript**
- Crafted a recursive algorithm to efficiently scan entire Markdown notes, noting the state of specific task items
- Accumulated an active user base, improving everyday productivity and enhancing the Markdown environment

Optiver Trading Bot <https://github.com/cyrusknopf/optibook-challenge/>

- Developed an algorithmic trading bot in Python for a mock financial exchange, accumulating **\$10,000/hr** at peak
- Leveraged arbitrage across **2 markets**, managing a **delta neutral** portfolio over 3 products and 2 baskets

Skills

C++, Nix, Linux CLI, Bash, Git, Valgrind, GDB, ARM Assembly, Python, Java, Rust, OCaml, Vim

Relevant Interests

Low latency C++, programming language design, lambda calculus, compilers, functional programming